WEAVE: Overcoming the Identity Shortcut in Style Transfer via 3-Minute Wavelet-Decoupled Flow Matching

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Abstract

Current style transfer methods face two primary challenges. First, state-of-the-art diffusion models, along with training-free methods built on them, require more than 60 GB of VRAM and therefore cannot run on consumer GPUs. Second, lightweight methods without these generative backbones often underperform a trivial identity mapping, which scores high on traditional metrics like ArtFID and LPIPS but was rarely evaluated as a baseline.

We empirically observe that frequency dominance in shared representation spaces pushes lightweight models toward identity-like solutions (Rahaman et al. 2019): low-frequency structural energy overwhelms high-frequency style gradients, a pattern we confirm through gradient-spectrum analysis and per-subband style-transfer measurements.

We argue that an effective style transfer method should outperform the identity baseline while remaining computationally affordable. To this end, we propose WEAVE: a single-level Haar wavelet transform splits the VAE latent into a low-frequency structure band and high-frequency style bands; a compact rectified-flow network learns a structure-preserving velocity field; and endpoint statistical alignment (AdaIN) injects the target style into the high-frequency subbands. This decoupling separates content transport from style injection, letting WEAVE train in 3 minutes on a single consumer GPU.

We also introduce an identity-floor-aware evaluation protocol, under which WEAVE clears the identity baseline with only 903K trainable parameters (plus a frozen 84M VAE), making high-quality style transfer both effective and affordable.

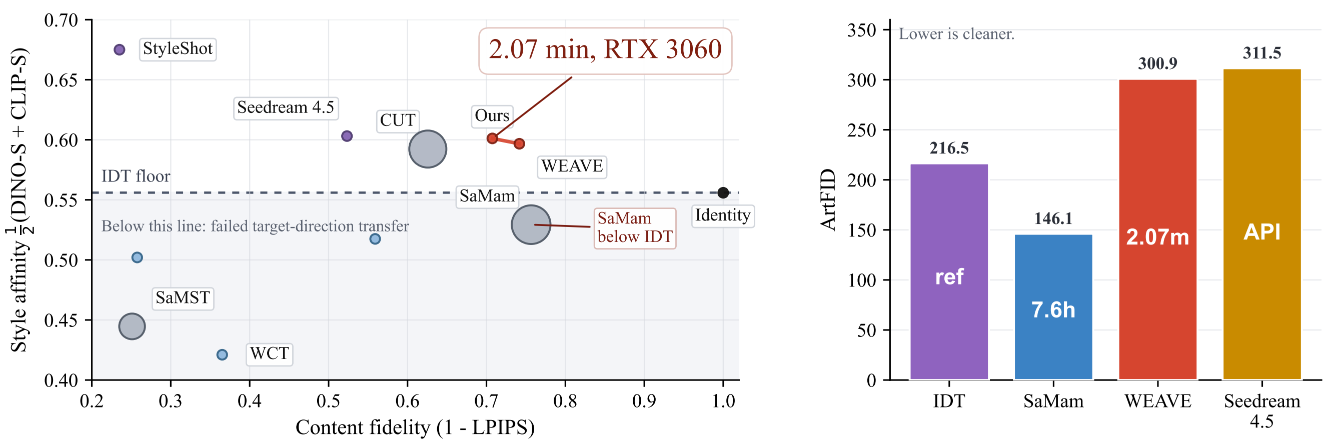

Figure 1: IDT-calibrated comparison on Distinct5-WikiArt. Left: style affinity $\frac{1}{2}(\mathrm{DINO\text{-}S} + \mathrm{CLIP\text{-}S})$ versus $1 - \mathrm{LPIPS}$, with bubble area encoding training time. The red points mark two WEAVE operating points. Square markers denote training-free diffusion editors (StyleAligned). The dashed line marks the IDT floor (0.556); methods below it have not moved in the requested direction. Right: target-pooled ArtFID on the same benchmark, with numbers inside the bars indicating training or response time.

Introduction

Style transfer faces two problems in deployment and evaluation. On one hand, state-of-the-art diffusion models (e.g., FLUX.2 and Qwen-image) and their training-free variants demand more than 60 GB of GPU memory, making them unusable on consumer hardware. On the other hand, certain lightweight alternatives that avoid these heavy generative backbones can struggle to achieve genuine style transfer. Specifically, the style affinity of outputs from certain lightweight models can fall below that of the unchanged source image (i.e., the identity mapping). This problem has been hidden by conventional evaluation protocols, such as LPIPS and

ArtFID, which heavily reward structural reconstruction but rarely include the identity mapping as an explicit baseline. As visualized in Figure 1 (Left), when the identity mapping is explicitly plotted, modern lightweight methods like SaMam clearly fall below this natural baseline. To fix this, we introduce the Identical-Image Transfer (IDT) calibration, transforming standard metrics into floor-calibrated measures.

Why do recent lightweight architectures, despite their advanced designs, consistently fall into this identity trap? We argue that a principal driver is not a lack of model capacity, but a well-documented spectral bias of neural networks (Rahaman et al. 2019; Xu, Zhang, and Luo 2022): in a shared representation space the optimizer preferentially fits the dominant low-frequency structure, a conflict that frequency-domain translation work has also linked to content–style entanglement (Cai et al. 2021). In typical flow-matching frameworks, the model predicts a velocity field that transports the source latent toward the target. However, natural images follow a $1/f$ spectral distribution, meaning the low-frequency structural energy (e.g., global layouts and object shapes) is orders of magnitude stronger than the high-frequency stylistic textures (e.g., brushstrokes and material patterns). When minimizing reconstruction loss in shared representation spaces, the optimizer naturally prioritizes fitting the massive low-frequency errors. Consequently, the subtle style-conditioned gradients are overwhelmed, pushing the model toward a style-agnostic identity mapping. This is why simply scaling up parameters can fail: structure and style compete in the same coordinate basis.

To solve the conflict between structure and style, we propose WEAVE. We operate in the latent space of a pre-trained diffusion VAE, which compresses images into a clean, low-dimensional space, but discard the heavy diffusion network and train only a tiny (903K-parameter) velocity network. A single-level Haar wavelet transform separates the latent into structure (LL) and texture (LH/HL/HH); style is injected only through the high-frequency subband statistics (endpoint AdaIN), which stay anchored to their original spatial layout by construction. The style-injection operator never touches the LL band, so content structure is preserved throughout the 8-step integration. This streamlined design lets WEAVE clear the identity baseline and train in 3 minutes on a single consumer GPU; because the alignment operates directly on latent features, it also serves as a training-free plugin for large pre-trained models, which we validate on a frozen SD1.5 image-to-image pipeline (Section).

Contributions. (1) **Evaluation Protocol:** We introduce the Identical-Image Transfer (IDT) calibration, exposing the blind spot where lightweight models underperform the unchanged source. (2) **Representation and Methodology:** We empirically observe that frequency dominance in shared representation spaces causes low-frequency structures to overshadow high-frequency textures, driving models toward the identity shortcut, and propose WEAVE, a wavelet-decoupled framework that achieves genuine style transfer with only 903K trainable parameters. Mechanism experiments (gradient-spectrum analysis, per-subband style-transfer ratios, and inference-time module swapping) provide causal evidence for this functional specialization. (3) **Efficiency and Generalization:** WEAVE trains in just 3 minutes on a single consumer GPU, and its latent operations are a pluggable, training-free mechanism for large diffusion backbones, which we validate by applying them to a frozen SD1.5 image-to-image pipeline (Section).

Related Work

Neural Style Transfer

Artistic style transfer was introduced by optimizing pixel spaces to match feature statistics of a pretrained network (Gatys, Ecker, and Bethge 2015, 2016), and was accelerated by matching feature statistics through normalization or transform layers, e.g., AdaIN (Huang and Belongie 2017) and whitening–coloring transforms (WCT) (Li et al. 2017). Subsequent GAN- and contrastive-learning approaches address unpaired translation, notably CUT (Park et al. 2020); later feed-forward and attention-based variants improve speed and fidelity (Park and Lee 2019; Deng et al. 2022; Hong et al. 2023; Kwon et al. 2024), while training-free directions reuse pretrained codebooks (Gim, Yoo, and Uh 2024) or hierarchical vision transformers (Zhang et al. 2024a), and diffusion backbones enable prompt-guided transfer (Zhang et al. 2024b). A recent survey frames a decade of progress and open gaps (Zhang and Tang 2025); however, these methods either need per-style optimization, risk content distortion when content and style share low-frequency statistics, or still demand substantial compute, leaving room for lightweight yet faithful transfer.

Diffusion-based and Training-free Style Editing

Latent diffusion models (Rombach et al. 2022) shifted style transfer from explicit feature transforms to generative editing in compressed latent spaces. Attention-control and inversion techniques enable training-free editing—Prompt-to-Prompt (Hertz et al. 2022) and SDEdit (Meng et al. 2021)—while attention-manipulation methods such as StyleID (Chung, Hyun, and Heo 2024) and StyleAligned (Hertz et al. 2024), image-prompt adapters like IP-Adapter (Ye et al. 2023), and inversion-based schemes like InST (Zhang et al. 2023) and DiffuseIT (Kim, Kwon, and Ye 2023) broaden the toolkit, with ArtBank (Zhang et al. 2024b) and Z-STAR (Lu et al. 2024) as recent training-free examples; yet state-of-the-art backbones (FLUX (Black Forest Labs 2024), Qwen-Image (Alibaba Cloud 2024), Seedream) and their editing variants rely on 60 GB+ VRAM and massive pretrained priors, limiting access on consumer hardware and motivating lightweight alternatives.

Flow Matching and Schrödinger Bridges for Image Translation

Transport-based formulations offer a principled alternative to adversarial translation. Rectified flow (Liu, Gong, and Liu 2023) and flow matching (Lipman et al. 2022) learn straight transport maps with few integration steps, while the Schrödinger bridge (SB) problem (Léonard 2014) and its diffusion/flow solvers—DSB (De Bortoli et al. 2021), Diffusion SB Matching (DSBM) (Shi et al. 2023), I2SB (Liu et al. 2023), and DDIB (Su et al. 2023)—couple transport to

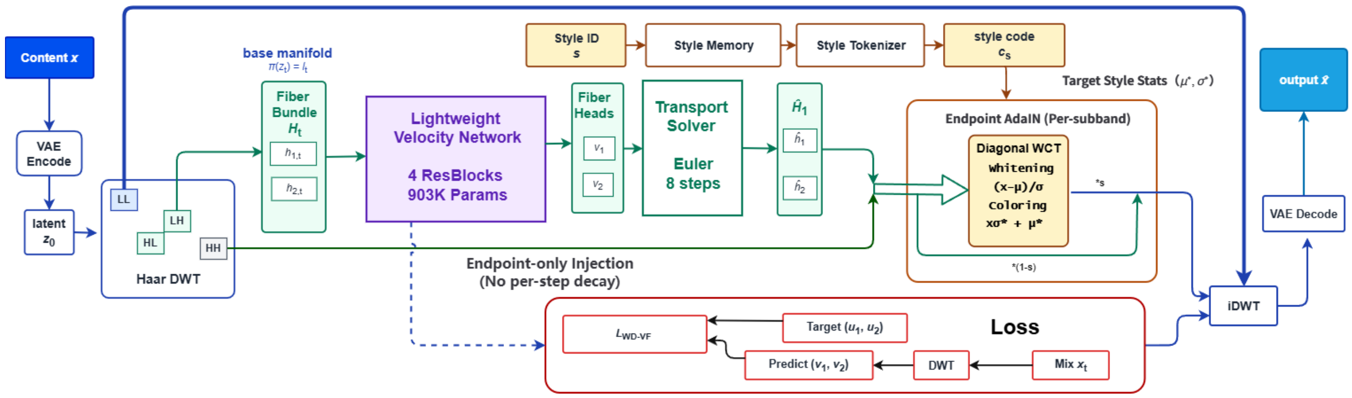

Figure 2: **WEAVE architecture**. A single-level Haar transform exposes the LL/LH/HL/HH subbands of the SD1.5 VAE latent. The compact velocity network v_θ (~903K) predicts motion for LL/LH/HL via three heads, while HH carries no head and is architecturally frozen. Training combines band-weighted rectified-flow matching (8-step Euler) with endpoint high-frequency statistical alignment (per-subband AdaIN), applied once at the final step on LH/HL/HH with scales $0.3/0.3/0.5$ and LL excluded. Style conditioning enters only the shared backbone (weak, λ_{LL}=0.3); a style-memory cross-attention branch is auxiliary.

optimal transport (Cuturi 2013; Peyré and Cuturi 2019) to map between distributions without paired data, but they learn a map in a shared coordinate basis where content and style compete, so they inherit the same low-frequency bias that can drive identity collapse in style transfer, and few exploit frequency structure to separate the two.

Lightweight Architectures and Evaluation Protocols

To reduce cost, recent lightweight designs train small networks on pixel or feature spaces, e.g., Mamba-based SaMST and SaMam (Liu et al. 2024, 2025; Gu and Dao 2023), yet standard protocols—LPIPS (Zhang et al. 2018) and ArtFID (Wright and Ommer 2022)—heavily reward source reconstruction and historically omit the identity mapping as an explicit baseline, so methods that merely preserve content can appear competitive; this motivates explicitly including the identity mapping as a direction-aware evaluation baseline.

Spectral Bias and Frequency-aware Optimization

Neural networks exhibit a well-documented spectral bias (Rahaman et al. 2019; Xu, Zhang, and Luo 2022), preferentially fitting low-frequency signals; together with the $1/f$ spectral distribution of natural images, this motivates frequency-aware optimization. Frequency-domain translation explicitly constrains low-/high-frequency statistics to protect content identity (Cai et al. 2021). Prior work applies spectral decomposition to restoration and generation, but the link between spectral bias and the identity shortcut in style transfer has not been drawn; we hypothesize that when content and style share one coordinate basis the optimizer's low-frequency preference suppresses style, motivating a decoupling of the two in the frequency domain.

Wavelet and Latent-space Alignment

Wavelet transforms separate multi-scale information in restoration and generation (Phung, Dao, and Tran 2023) and have been adapted to style transfer (Yoo et al. 2019); the Haar basis in particular gives exact orthogonality and additive norm splitting (Daubechies 1992). Concurrently, latent diffusion (Rombach et al. 2022) yields regularized latent spaces where statistical alignment such as WCT (Li et al. 2017) is stable.

Motivation. Combining wavelet decomposition with latent-space statistical alignment inside a flow-matching transport—structurally separating content from style rather than competing in a shared space—directly addresses the limitations above and motivates the WEAVE framework in Section .

Method

We describe WEAVE following the logic of the Introduction: first, where to learn the transfer (VAE latent space); second, how to separate conflicting signals (wavelet decomposition); third, how to route style information while protecting content (frequency-aware architecture and endpoint AdaIN); and fourth, how the learned transport and the statistical aligner cooperate at inference.

Framework Setup

Let $x_c \in \mathbb{R}^{H\times W\times 3}$ be a content image and $s \in \{1,\ldots,K\}$ a target style domain. We operate in the latent space of the Stable Diffusion v1.5 EMA VAE (Rombach et al. 2022), writing $z_0 = \mathcal{E}(x_c) \in \mathbb{R}^{4\times 32\times 32}$; during training $z_1 \sim p_s$ is sampled from the target domain. Rectified flow parameterizes the straight transport segment

$$z_t = (1-t)z_0 + tz_1, \quad u_t = z_1 - z_0, \quad t \sim \mathcal{U}([0,1]), \tag{1}$$

whose straight trajectory needs only 8 Euler steps (vs. thousands of diffusion denoising steps), keeping training and inference on a single consumer GPU (inference details in Section).

Wavelet Decomposition and Frequency Routing

As discussed in Section 2.3, natural images follow a $1/f$ spectral distribution: low-frequency structural energy dom-

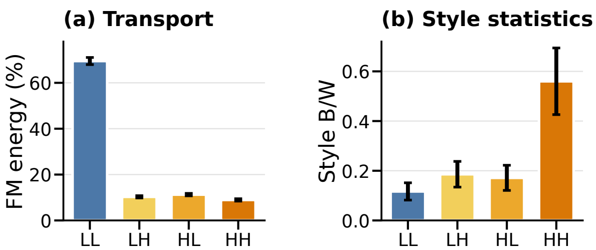

Figure 3: **Frequency probes.** (a) Cross-style transport energy in each Haar band. (b) Between-style versus within-style separation of latent statistics. Error bars summarize variation across style routes or statistics.

inates over high-frequency textures (Rahaman et al. 2019; Xu, Zhang, and Luo 2022). Learning style transfer in shared representation spaces biases the optimizer toward structure over style (a known failure mode in frequency-domain translation (Cai et al. 2021)). WEAVE addresses this by changing coordinates.

Haar wavelet decomposition. A single-level Haar wavelet transform decomposes the $4\times32\times32$ latent into four orthogonal subbands: a low-frequency LL band and three high-frequency bands (LH, HL, HH), each of size $4\times16\times16$. We write $\mathcal{W}(z_t) = (\ell_t, h_{1,t}, h_{2,t}, h_{3,t})$ and $\mathcal{W}(u_t) = (u_\ell, u_1, u_2, u_3)$. The transform is orthogonal, so the Frobenius norm splits additively:

$$\|z\|_F^2 = \|\ell\|_F^2 + \|h_1\|_F^2 + \|h_2\|_F^2 + \|h_3\|_F^2. \tag{2}$$

Due to the orthogonality of the Haar transform (Daubechies 1992), modifying only the high-frequency subbands leaves the LL coordinate unchanged—structure is preserved by construction. Haar is selected for its exact orthogonality, compact spatial support, and very low cost; a single decomposition level suffices for the compact latent resolution.

Empirical frequency probes. Across 5,000 random cross-style latent pairs, transport energy concentrates in LL while style separation is strongest in HH (Figure 3), motivating weak LL supervision and stronger high-frequency emphasis.

Architecture. The velocity network consists of four residual blocks (width 64, $\sim$903K trainable parameters) with three separate prediction heads—one 1×1 convolution per supervised subband (ℓ, h_1, h_2). The h_3 (HH) band receives no velocity head, so the Euler integrator never modifies it; it is stylized only at the final (endpoint) step via the per-subband AdaIN operator (Section). The LL subband enters the backbone as a separate channel and interacts with style only through the shared residual blocks—never through the attention path—preserving the intended frequency separation. We use RMSNorm throughout, as GroupNorm erases the channel-wise mean carrying tonal information. The overall architecture is shown in Figure 2.

Spectral flow-matching objective. The spectral flow-matching term is

$$\mathcal{L}_{\text{impl}} := \mathbb{E}_t\left[\lambda_{LL}\|v_\ell - u_\ell\|_2^2 + \|v_{h_1} - u_{h_1}\|_2^2 + \|v_{h_2} - u_{h_2}\|_2^2\right], \tag{3}$$

with $\lambda_{LL} = 0.3$. WEAVE de-weights LL rather than eliminating it, keeping a weak anchor because some styles also shift coarse color tone; ablation confirms this choice is pivotal (Table 2). The predicted one-step target is $\hat{z}_1 = z_0 + \mathcal{W}^{-1}(v_\ell, v_{h_1}, v_{h_2}, 0)$.

Mild LL partial stylization. While LL is structurally locked to preserve content, a mild partial stylization of the LL *target* lets coarse color statistics migrate toward the style domain without sacrificing structure:

$$\ell_1^{\text{target}} = (1-\alpha)\,\ell_0 + \alpha\,\text{AdaIN}(\ell_0, \ell_1^\star), \qquad \alpha = 0.3, \tag{4}$$

where ℓ_0 is the source and $\ell_1^\star$ a style-reference LL subband. We set α=0.3 because aggressive blends ($\alpha\geq$0.5) damage content without raising DINO-S (Table 2).

Endpoint AdaIN: Primary Style Injection Channel

The flow-matching velocity field transports the content latent toward the target domain, but the velocity signal alone is a weak style channel—the LL-dominated reconstruction loss leaves little room for style-conditioned gradients. WEAVE therefore injects the target style through a per-subband statistics operator applied once at the final (endpoint) integration step. We restrict alignment to the endpoint because applying it at intermediate Euler steps would re-inject target statistics into the high-frequency subbands at every step and, via the shared backbone, leak target structure into the LL band—causing the low-frequency content to drift; confining it to the final step keeps the 8-step transport purely content-preserving and adds style only as a final recolor. We call this *AdaIN*, and our ablation confirms it is the single most critical component: removing it reverts the output to content preservation (Table 2).

Alignment operator. Let $\mathcal{W}(\hat{z}_1) = (\hat{\ell}, \hat{h}_1, \hat{h}_2, \hat{h}_3)$ and $\mathcal{W}(z_1^\star) = (\ell^\star, h_1^\star, h_2^\star, h_3^\star)$ be the subbands of the current prediction and a style reference sampled from the target domain. For $i \in \{1,2,3\}$, the per-subband AdaIN map (Huang and Belongie 2017) matches channel-wise statistics:

$$T_i(a) = \frac{a - \hat{\mu}_i}{\hat{\sigma}_i}\,\sigma_i^\star + \mu_i^\star, \tag{5}$$

where $(\hat{\mu}_i, \hat{\sigma}_i)$ and $(\mu_i^\star, \sigma_i^\star)$ are the per-channel mean and standard deviation of $\hat{h}_i$ and $h_i^\star$. Because each Haar subband has only four channels, the full-covariance whitening–coloring transform collapses to this diagonal (mean+std) form (Li et al. 2017). With high-frequency scales $(s_1, s_2, s_3) = (0.3, 0.3, 0.5)$, the aligned subbands are

$$\hat{h}_i^+ = (1-s_i)\hat{h}_i + s_i\,T_i(\hat{h}_i), \qquad i \in \{1,2,3\}, \tag{6}$$

and we set $\mathcal{W}(\hat{z}_1^+) = (\hat{\ell}, \hat{h}_1^+, \hat{h}_2^+, \hat{h}_3^+)$ with $\hat{\ell}$ unchanged. The alignment is applied once at the final (endpoint) Euler step. The LL structure is preserved by construction; the VAE's KL regularization makes the latent approximately Gaussian, justifying second-order moment matching.

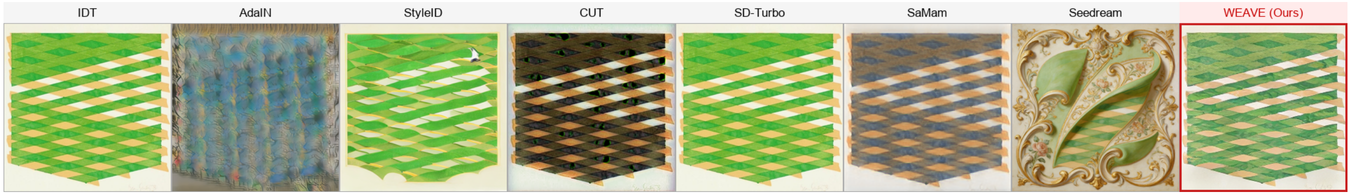

Figure 4: Extreme cross-domain stylization (Minimalism → Rococo). Heavy diffusion priors (Seedream, StyleID) hallucinate content, while classical methods and SaMam collapse to near-identity. WEAVE preserves source topology via low-frequency anchoring and injects the target's painterly textures exclusively through high-frequency subbands.

Training Objective

The training loss is

$$\mathcal{L} = \mathcal{L}_{\text{impl}} + \underbrace{0.1\mathcal{L}_{\text{edge}} + \mathcal{L}_{\text{low}}}_{\text{auxiliary } (<5\% \text{ gradient})}, \tag{7}$$

where $\mathcal{L}_{\text{edge}}$ is an ℓ_1 match between high-pass residuals and $\mathcal{L}_{\text{low}}$ is an MSE anchor on the low-pass component. The spectral flow-matching term $\mathcal{L}_{\text{impl}}$ alone drives learning ($\sim$92% of the gradient); the auxiliary terms contribute $< 5\%$ combined, and their individual removal does not measurably change the output. Ablation confirms $\mathcal{L}_{\text{impl}}$ is the engine of content-to-style transport—removing it improves content metrics but drops style (Table 2). WEAVE trains on a single consumer GPU; full training hyperparameters are in the supplementary material. The learning rate is the only sensitive training hyperparameter; all other knobs are robust.

Inference Configuration

At inference we integrate from z_0 to $\hat{z}_1$ with an 8-step Euler solver. The step count is the most consequential inference knob: a single step collapses, while 32 steps matches 8—eight is the efficiency–accuracy Pareto point. A style-extrapolation coefficient α_{extrap}=0.1 pushes the aligned statistics slightly beyond the target reference; α=1.0 causes catastrophic collapse as the matched covariance becomes singular.

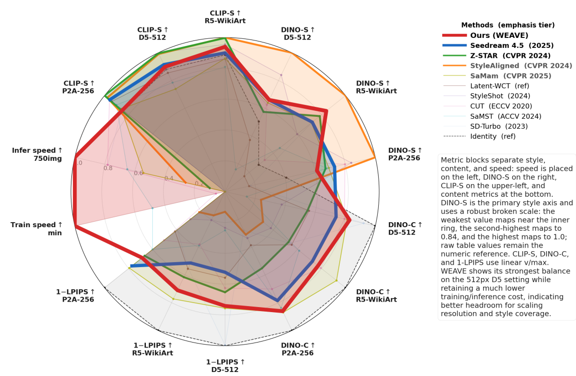

Figure 5: Multi-metric radar profile across D5-512, P2A-256, and R5-WikiArt. Axes are grouped by role: speed on the left, DINO-S on the right, CLIP-S on the upper-left, and content metrics at the bottom. DINO-S, our primary style metric, uses a robust broken visual scale (minimum near the inner ring, second-highest at 0.84, maximum at the outer ring) to expose both the low-style Latent-WCT baseline and the middle tier; this scaling is visual only, and Table 1 remains the source for raw values. CLIP-S, DINO-C, and 1-LPIPS use linear per-axis $v/\max$. Line weight encodes emphasis tier rather than rank. WEAVE (red) gives the strongest content-preserving profile among learned methods, especially on D5-512, while keeping much lower training and inference cost, suggesting stronger scaling potential at higher resolution and broader style coverage.

Experiments

Setup and Protocol

Setup. We use Distinct5-WikiArt-512 with five styles (Early Renaissance, Impressionism, Minimalism, Rococo, Ukiyo-e) chosen for their low IDT CLIP-S, making accidental no-op success harder. Each domain has 3,600 training and 150 test images; evaluation uses 750 source-target pairs including 150 identity pairs for IDT calibration, and generalization is tested on all 20 WikiArt families (20×20×30=12,000 pairs) with baselines confined to the Distinct5 subset (750 pairs) to bound compute. **DINO-S** (DINOv2-small CLS-token cosine similarity to the target-style reference pool) is our primary style metric, with CLIP-S (ViT-B/32) secondary; LPIPS (AlexNet) and DINO-C measure content preservation. Baselines: Identity, SD-Turbo, StyleAligned, Z-STAR, StyleShot, CUT, SaMST, SaMam, Seedream 4.5, and Latent-WCT.

Training and evaluation. WEAVE is a compact network trained on a single consumer GPU; architecture and training hyperparameters are in the supplementary material. All methods are evaluated on every ordered source-target pair; learned baselines are selected by best validation CLIP-S on a 50-pair held-out split, and all metrics use deterministic seeds.

Main Results

The IDT protocol redefines success. The identity row turns CLIP-S into a direction-aware metric: once included, low LPIPS alone is not enough, and both SaMST and SaMam stay below the IDT floor, suggesting their low LPIPS reflects source preservation.

WEAVE occupies the practical frontier. On D5-512, the only methods matching or exceeding WEAVE on DINO-S all pay a severe content price, with much higher LPIPS and lower DINO-C (Table 1); WEAVE thus sits at the content-preserving end of the Pareto frontier, already near this structural ceiling.

Method	D5-512				P2A-256				R5-WikiArt				Params	Train	Infer
	DINO-S↑	CLIP-S↑	LPIPS↓	DINO-C↑	DINO-S↑	CLIP-S↑	LPIPS↓	DINO-C↑	DINO-S↑	CLIP-S↑	LPIPS↓	DINO-C↑		min	750 imgs
Identity	0.419	0.693	0.000	1.000	0.416	0.663	0.000	1.000	0.433	0.731	0.000	1.000	0	free	0 s
SD-Turbo	0.484	0.693	0.003	0.922	0.341	0.674	0.603	0.279	0.505	0.767	0.449	0.538	$0^{+1.1B}$	free	5 m
StyleAligned	0.675	0.780	0.869	0.239	0.612	0.768	0.786	0.310	0.649	0.824	0.829	0.315	$0^{+1.1B}$	free	77 m
Z-STAR	0.449	0.784	0.347	0.549	0.514	0.786	0.332	0.526	0.514	0.822	0.384	0.526	$0^{+1.1B}$	free	3 h
StyleShot	0.563	0.787	0.765	0.377	0.552	0.774	0.713	0.335	0.484	0.787	0.795	0.380	$0^{+2.4B}$	free	5.1 h
CUT	0.471	0.714	0.374	0.795	0.539	0.754	0.494	0.702	0.441	0.710	0.620	0.795	7.0M	322.6	5 m
SaMST	0.271	0.618	0.749	0.145	0.501	0.709	0.398	0.838	0.461	0.667	0.612	0.615	8.3M	39.5	10 m
SaMam	0.477	0.582	$\underline{0.243}$	0.812	0.505	0.677	$\underline{0.205}$	0.870	0.503	0.712	$\underline{0.227}$	0.925	8.8M	436.0	17.6 m
Seedream 4.5	0.486	0.720	0.477	0.739	0.517	0.752	0.227	0.785	0.504	0.742	0.486	0.725	–	API	–
Latent-WCT	0.362	0.673	0.441	0.559	0.338	0.738	0.585	0.386	0.414	0.710	0.444	0.553	0	free	18 s
Ours	**0.4859**	0.7075	**0.2583**	**0.8287**	0.4801	0.6681	0.3116	0.8612	$\underline{0.5226}$	0.7747	0.2895	0.7717	**903K**$^{+84M}$	**3.0**	**95 s**

Table 1: Main comparison on D5-512, P2A-256, and R5-WikiArt. CLIP-S, DINO-S, and DINO-C are higher better; LPIPS is lower better. DINO-S is the primary style metric (see Section). Superscripts in Params denote frozen backbones, and "–" denotes withheld or unavailable values. Infer is single-RTX-3060 (12 GB) generation-only wall-clock for 750 pairs (8-step ODE + VAE decode; metrics excluded).

Among trained methods that clear the IDT floor, WEAVE also achieves the lowest LPIPS and the only sub-million-parameter profile; CUT needs an order of magnitude more parameters and training time to do the same (Table 1). The full multi-metric profile across all three benchmarks and both speed axes is summarized in Figure 5.

Generalization across 20 families. Because Distinct5 uses only five hard, low-IDT styles, we also train and evaluate on all 20 WikiArt families (R5-WikiArt, 12,000 pairs): WEAVE clears the IDT floor here too (Table 1), with CLIP-S rising and LPIPS essentially unchanged despite a $16\times$ larger style space. On P2A-256 it still clears the floor—expected, since lower resolution compresses the high-frequency subbands and the larger style space raises the floor—while avoiding negative transfer. Because the high-frequency path is decoupled from content transport, a new style can be added from a few references (30 suffice) by updating only the style-memory embedding, without retraining the backbone.

Empirical Efficiency. The Train column of Table 1 shows that backbone-free methods are lightweight only at inference, underscoring that low inference cost does not imply low training cost. Our simplified wavelet architecture instead trains in 3 minutes on a single RTX 3060 and generates all 750 pairs in 95 s (generation-only; VAE decode $\approx$42% of that wall-clock): confining supervised transport to the high-frequency subbands shrinks the parameter space the optimizer must explore (the dominant-energy LL band never enters the style-conditioned loss), leaving VAE decode as the dominant cost.

Mechanistic diagnosis and comparison. As the frequency-dominance analysis predicts, SaMam shows the style-suppression pattern (sub-IDT CLIP-S despite low LPIPS), while iterative editors such as StyleID suffer per-step content decay; WEAVE's endpoint high-frequency AdaIN avoids this decay, reaching comparable target-direction transfer with far less content damage. Relative to Seedream 4.5, WEAVE reaches comparable style with far lower LPIPS, under one million parameters, on a single consumer GPU. The comparison with Latent-WCT confirms that wavelet decomposition alone, without learned transport, is insufficient (Table 1).

Auxiliary metrics corroborate the main findings. Figure 1 (right) shows WEAVE's target-pooled ArtFID is lower than Seedream 4.5's; on all-pairs evaluation WEAVE reaches competitive CLIP-S with the lowest LPIPS. The DINO-based metrics corroborate this: WEAVE's high DINO-C (near the identity ceiling) and DINO-S near the structural ceiling confirm genuine style acquisition rather than mere content preservation.

Mechanistic Ablations

Table 2 lists all ablations with a measurable effect on D5-512; many hyperparameter perturbations ($10\times$–$20\times$ on σ, g_0, λ_{HH}, MSE$\rightarrow$Huber) move CLIP-S by < 0.002 and are omitted. AdaIN is the most critical component (removing it reverts output to content preservation), Flow Matching drives transport (removing it improves content but drops style), and step count / extrapolation show clear collapse points.

Config	CLIP-S↑	LPIPS↓	DINO-S↑	DINO-C↑
Baseline	**0.727**	**0.343**	**0.483**	**0.755**
w/o AdaIN	0.708	0.334	0.484	0.816
w/o Flow Matching	0.710	0.296	0.475	0.819
w/o Cross-attention	0.726	0.336	0.486	0.765
λ_{LL}=0	0.713	0.313	0.480	0.804
lr $= 5\times10^{-4}$	0.730	0.356	0.487	0.738
AdaIN α=0	0.709	0.339	0.483	0.811
AdaIN α=2.0	0.717	0.297	0.485	0.802
extrap $= 1.0$	0.704	0.591	0.411	0.616
num_steps $= 1$	0.709	0.476	0.376	0.448

Table 2: Ablation on D5-512 (5-epoch baseline, 750 test pairs). The main-table configuration (10 epochs, α=0.3 LL partial stylization) improves all four metrics; see Section . Configurations with ΔCLIP-S< 0.002 are omitted.

Latent-WCT baseline. Replacing learned transport with a single-shot covariance match (zero parameters) collapses on all four metrics (Table 1), confirming the 8-step Rectified Flow—not the wavelet operations—is the key mechanism.

Style ceiling. Ten variants strengthening the style pathway all converge to DINO-S ≈ 0.48 under a 5-epoch budget, confirming this as the structural ceiling of the structure-aligned-target paradigm at 903K parameters. The main configuration (α=0.3 LL blend, 10 epochs) breaks it as a Pareto improvement, since only a mild blend lets color/contrast statistics migrate into the LL target without overwhelming the content anchor; fully unlocking LL would leave the paradigm.

Summary. WEAVE's behavior is governed by three principal components (AdaIN, Flow Matching, ODE step count) and one sensitive hyperparameter (learning rate); the rest are robust. Causal evidence follows in Section .

Generalization and Robustness

Latent space vs. RGB. A matched 256×256 control isolates the source of the gain: at equal training time and parameters, direct RGB flow reaches far lower CLIP-S with much worse LPIPS, confirming the gain comes from latent-space wavelet transport, not an extra loss.

Few-shot adaptation. Freezing the backbone and updating only the 30-image style memory clears the IDT floor on all 8 held-out styles; fine-tuning adds only a small CLIP-S gain at higher cost.

Zero-shot transfer. We take the *D5-trained* checkpoint (5 styles) and evaluate it on Other5—five WikiArt families unseen during training (750 pairs). Table 3 shows WEAVE still clears the identity floor and wins on content preservation: its LPIPS is far below SaMam and SaMST while its CLIP-S and DINO-C match the best; SaMam only edges WEAVE on DINO-S, trading brushstroke matching for content fidelity.

Method	CLIP-S ↑	LPIPS ↓	DINO-S ↑	DINO-C ↑
Ours (WEAVE)	**0.7742**	**0.2802**	0.5313	**0.7587**
SaMam	0.7577	0.3990	**0.5466**	0.7369
SaMST	0.7485	0.6213	0.3913	0.2680

Table 3: Zero-shot evaluation on Other5 (all 750 source→target pairs). The D5-trained checkpoint transfers to 5 unseen WikiArt families, halving LPIPS versus SaMST and beating SaMam by ∼30% while matching the best CLIP-S and DINO-C.

Wavelet basis and resolution. Daubechies-2 slightly raises CLIP-S but worsens LPIPS, so we retain Haar for its exact orthogonality and local support; native-resolution training benefits the high-frequency bands, whereas upsampling a low-resolution model is far worse.

Plug-and-Play Extension to Pre-trained Diffusion Models

To show WEAVE's latent operations are pluggable rather than tied to our 903K network, we apply the same wavelet decoupling and high-frequency statistical alignment (AdaIN) to a frozen SD1.5 image-to-image pipeline without training any parameters: after empty-prompt I2I (strength 0.5, 20-step DDIM), a single-level Haar DWT matches the high-frequency subbands (LH/HL/HH) to the target style (AdaIN, scale 0.8) while keeping LL unchanged. On the off-diagonal D5 pairs (n=600), this raises CLIP-S and DINO-S significantly (paired t-test $p<10^{-15}$) while barely touching content; the gains hold across all 750 pairs. At 4.9 ms per image, this analytic DWT+AdaIN+IDWT confirms WEAVE plugs into pre-trained diffusion backbones with almost no content damage.

Mechanism Analysis

Gradient-spectrum evidence for frequency dominance. We provide evidence that the identity shortcut is strongly associated with—and plausibly driven by—frequency dominance, by measuring the gradient spectrum during training. After one forward pass at t=0.5, the unweighted low-frequency loss is $5\times$ larger than the high-frequency losses, and the LL head's gradient norm is $6\times$ larger than the other heads. Even with explicit down-weighting, the LL input-gradient energy share remains the largest single-subband share, confirming that low-frequency dominance is structural rather than a loss-weighting artifact.

Per-subband functional specialization. Decomposing the output latent and measuring each subband's statistical distance to the target style reveals distinct functional roles: mid-frequency subbands (LH/HL) are the primary style carriers, LL preserves content structure, and HH is architecturally frozen—the model has no head$_{HH}$, so the Euler integrator never modifies the finest diagonal detail. Wavelet decoupling is thus not merely a training convenience but an enforced structural partition.

Discussion

Why wavelet decoupling works. The mechanism analysis confirms frequency dominance is structural—even down-weighted LL still holds the largest single-subband gradient share—and that wavelet decoupling enforces a hard partition (LL content, LH/HL style, HH frozen), turning a soft optimization conflict into a structural one; this is why endpoint AdaIN, the primary style-injection path that bypasses the LL-dominated reconstruction loss, is the most critical component.

Flow Matching and AdaIN cooperate, and AdaIN is the right operator. Flow Matching plans a content-preserving trajectory, while endpoint AdaIN performs exact high-frequency alignment; the two combine synergistically beyond either alone (Table 2). The 8-step integration lets each operator correct the other's drift. An inference-time module swap confirms AdaIN is the primary, not redundant, injection channel: disabling it collapses the LH style-transfer ratio, while full-covariance WCT helps only marginally at the cost of content drift, and per-subband decomposition is strictly worse because the model was trained with the global spatial-fiber mode.

Limitations. WEAVE is domain-conditional (a style family, not a reference painting). The Distinct5 benchmark is intentionally narrow (five hard, low-IDT styles as a no-op stress test), partially addressed by the Random5-WikiArt extension; the LL-protecting design makes palette-heavy targets like Minimalism harder than textured ones. The single-level Haar decomposition may be suboptimal for styles whose distinguishing features span multiple scales; multi-level extensions are left for future work.

Future directions. The wavelet-decoupled framework naturally extends to multi-level decompositions that separate palette, layout, and brushstroke into independent channels, and to reference-image conditioning that bridges domain-level and arbitrary-image style transfer while preserving the efficiency of frequency routing.

Conclusion

Style transfer needs both the right success criterion and the right coordinate system. IDT calibration exposes when a method merely reconstructs the source, and WEAVE avoids this regime by separating low-frequency structure from high-frequency style motion and routing style through the same high-frequency path used at inference. The 903K-parameter backbone trains in 3 minutes on one RTX 3060, with 95 s generation-only inference for 750 pairs. The plug-and-play extension to SD1.5 demonstrates that the wavelet-decoupled mechanism generalizes beyond our compact network to large pre-trained diffusion backbones. The central finding: real target-direction transfer can be achieved without a large model once the transport geometry separates structure from style, and by running on a single consumer GPU WEAVE makes high-quality stylization accessible.

Ethics Statement

Style transfer tools can be misused to produce deceptive images or to misattribute stylistic authorship. Our method operates at the domain level (e.g., "Impressionism") rather than the per-artist level. This reduces the risk of imitating a specific living artist. We do not condition on individual reference paintings. The dataset is sourced from WikiArt and contains historically significant artworks. It does not contain personal data. We encourage downstream users to disclose when an image has been style-transferred and to respect source-content licensing terms.

Reproducibility Statement

All experiments use public data (WikiArt) and a public VAE (Stable Diffusion v1.5 EMA VAE). The experiments section gives the training configuration, hyperparameters, and evaluation protocol. Training takes 3 minutes on a single RTX 3060 (12 GB; 10-epoch wall-clock, batch 96). Generation-only inference for 750 pairs needs about 95 seconds (8-step ODE + VAE decode; $\approx$71 ms network generation and $\approx$52 ms decode per image), and a full metrics pass (CLIP/LPIPS on top of generation) is about 2 minutes. The 903K trainable parameter backbone checkpoint is 3.5 MB. We will release the code, the checkpoint, and the 750-pair evaluation protocol upon publication.

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